

Task 1: Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that

1. Accept a Linux command as input.
2. Create a child process using `fork()`.
3. Execute the command in the child process using an appropriate `exec()` system call.
4. Allow the parent process to wait for the child using `wait()`.
5. Display the Process ID (PID) of both parent and child processes.

- a) Write a C code to create a process using `fork()` system call.
- b) Read the user input as shell command.
- c) Use a system call `exec()` to execute the shell command.
- d) Use a `wait()` system call for waiting a parent process.
- e) Display the PIDs of parent and child processes.

```
GNU nano 7.2 p2t1.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;
    char command[100];

    // b) Read user input as shell command
    printf("Enter a Linux command: ");
    scanf("%s", command);

    // a) Create a child process using fork()
    pid = fork();

    if (pid < 0)
    {
        printf("Fork failed!\n");
        exit(1);
    }

    else if (pid == 0)
    {
        // Child Process
        printf("\nChild Process\n");

        // e) Display child and parent PIDs
        printf("Child PID : %d\n", getpid());
        printf("Parent PID : %d\n", getppid());

        printf("Executing command: %s\n", command);

        // c) Execute command using exec()
        execlp(command, command, (char *)NULL);

        // This executes only if exec() fails
        printf("Exec failed!\n");
        exit(1);
    }

    else
    {
        // Parent Process
        printf("\nParent Process\n");

        // e) Display parent and child PIDs
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        // d) Parent waits for child
        wait(NULL);

        printf("\nChild process completed.\n");
    }

    return 0;
}
```

```

satwika@SATWIKAG:~$ nano p2t1.c
satwika@SATWIKAG:~$ gcc p2t1.c -o p2t1
satwika@SATWIKAG:~$ ./p2t1
Enter a Linux command: ls

Parent Process
Parent PID : 1379
Child PID : 1382

Child Process
Child PID : 1382
Parent PID : 1379
Executing command: ls
fork.c fork2 fork2.c fork_exec fork_exec.c frok_exec helloworld myShell myShell.c p2t1 p2t1.c practical3.c process process.c states states.c task1 task1.c

Child process completed.
satwika@SATWIKAG:~$

```

Task-2: Using Linux terminal commands (uname, lscpu, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

```

Child process completed.
satwika@SATWIKAG:~$ gcc p2t1.c -o p2t1
satwika@SATWIKAG:~$ ./p2t1
Enter a Linux command: uname

Parent Process
Parent PID : 1481
Child PID : 1482

Child Process
Child PID : 1482
Parent PID : 1481
Executing command: uname
Linux

Child process completed.
satwika@SATWIKAG:~$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 16
On-line CPU(s) list:   0-15
Vendor ID:              GenuineIntel
Model name:             Intel(R) Core(TM) 7 240H
CPU family:             6
Model:                  186
Thread(s) per core:     2
Core(s) per socket:     8
Socket(s):              1
Stepping:               2
BogoMIPS:               5990.42
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc rep_good nopl xtopology tsc_relia
ble nonstop_tsc cpuid tsc_known_freq pni pclmuldq vmx ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3d
nowprefetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt clwb sha_ni xsaveopt xsav
ec xgetbv1 xsaves avx_vnni vnni umip waitpkg gfni vaes vpclmuldq rdpid movdiri movdir64b fsrm md_clear serialize ibt flush_l1d arch_capabilities

Virtualization features:
Virtualization:         VT-x
Hypervisor vendor:      Microsoft
Virtualization type:    full
Caches (sum of all):
L1d:                     384 KiB (8 instances)
L1i:                     256 KiB (8 instances)
L2:                      10 MiB (8 instances)
L3:                      24 MiB (1 instance)
NUMA:
NUMA node(s):            1
NUMA node0 CPU(s):      0-15
Vulnerabilities:
Gather data sampling:    Not affected
Ghostwrite:              Not affected
Indirect target selection: Not affected
Itlb multihit:           Not affected
L1tf:                    Not affected
Mds:                     Not affected
Meltdown:                 Not affected
Mmio stale data:         Not affected
Old microcode:           Not affected
Reg file data sampling:  Mitigation; Clear Register File
Retbleed:                 Mitigation; Enhanced IBRS
Spec rstack overflow:    Not affected
Spec store bypass:       Mitigation; Speculative Store Bypass disabled via prctl

```

```

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Meltdown: Not affected
Mmio stale data: Not affected
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Spec rstack overflow: Not affected
Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1: Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2: Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_DIS_S
Srbds: Not affected
Tsa: Not affected
Tsx async abort: Not affected
Vmscape: Not affected

```

```
satwika@SATWIKAG:~$ ps
```

```

  PID TTY          TIME CMD
  320 pts/0    00:00:00 bash
 1406 pts/0    00:00:00 ps

```

```
satwika@SATWIKAG:~$ top
```

```

top - 13:05:03 up 53 min, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 22 total, 1 running, 21 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.6 us, 0.0 sy, 0.0 ni, 99.4 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7787.6 total, 7080.9 free, 611.2 used, 251.7 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7176.4 avail Mem

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	21880	13176	9800	S	0.0	0.2	0:02.84	systemd
2	root	20	0	3180	2200	2072	S	0.0	0.0	0:00.01	init-systemd(Ub
6	root	20	0	3196	2128	2008	S	0.0	0.0	0:00.00	init
55	root	19	-1	66840	19628	18500	S	0.0	0.2	0:00.61	systemd-journal
112	root	20	0	25612	7056	5176	S	0.0	0.1	0:01.69	systemd-udevd
119	systemd+	20	0	21344	13212	11088	S	0.0	0.2	0:00.21	systemd-resolve
120	systemd+	20	0	91036	7992	7024	S	0.0	0.1	0:00.31	systemd-timesyn
189	root	20	0	4244	2684	2432	S	0.0	0.0	0:00.01	cron
190	message+	20	0	9672	5460	4736	S	0.0	0.1	0:00.25	dbus-daemon
207	root	20	0	17980	8856	7812	S	0.0	0.1	0:00.19	systemd-logind
224	syslog	20	0	222516	5936	4552	S	0.0	0.1	0:00.12	rsyslogd
231	root	20	0	3124	1972	1836	S	0.0	0.0	0:00.01	agetty
244	root	20	0	107036	23216	13764	S	0.0	0.3	0:00.13	unattended-upgr
313	root	20	0	3188	1104	980	S	0.0	0.0	0:00.00	SessionLeader
314	root	20	0	3204	1244	1108	S	0.0	0.0	0:00.10	Relay(320)
320	satwika	20	0	6080	5384	3664	S	0.0	0.1	0:00.08	bash
324	root	20	0	6704	4660	3876	S	0.0	0.1	0:00.00	login
384	satwika	20	0	20320	11584	9456	S	0.0	0.1	0:00.18	systemd
385	satwika	20	0	21168	3596	1868	S	0.0	0.0	0:00.00	(sd-pam)
411	satwika	20	0	6080	5336	3668	S	0.0	0.1	0:00.03	bash
1055	polkitd	20	0	308172	8108	7212	S	0.0	0.1	0:00.08	polkitd
1407	satwika	20	0	9344	5464	3244	R	0.0	0.1	0:00.00	top